When Can You Poison Rewards?

A Tight Characterization of Reward Poisoning in Linear MDPs

Oregon State University
ILLINOIS
PRINCETON UNIVERSITY
BOSTON UNIVERSITY
ICML International Conference On Machine Learning

Jose E. Aguilar Escamilla Haoyang Hong Jiawei Li Haoyu Zhao
Xuezhou Zhang Sanghyun Hong Huazheng Wang

View our Paper:

Limits of Reward Poisoning

Reinforcement learning's (RL) success in problems like robotics, LLM training/alignment, and healthcare, depends on the quality of the reward feedback. Trustworthy research has shown a high potential for malicious manipulation through reward poisoning. Our paper explores the fundamental limits of reward poisoning by studying Intrinsic Robustness:

"Can the 'intrinsic structure' of a learning problem make certain adversarial objectives impossible?"

Our main contribution, a *convex quadratic program* (CQP test) capturing the conditions for intrinsic robustness, which we prove and demonstrate empirically.

Theoretical & Threat Models

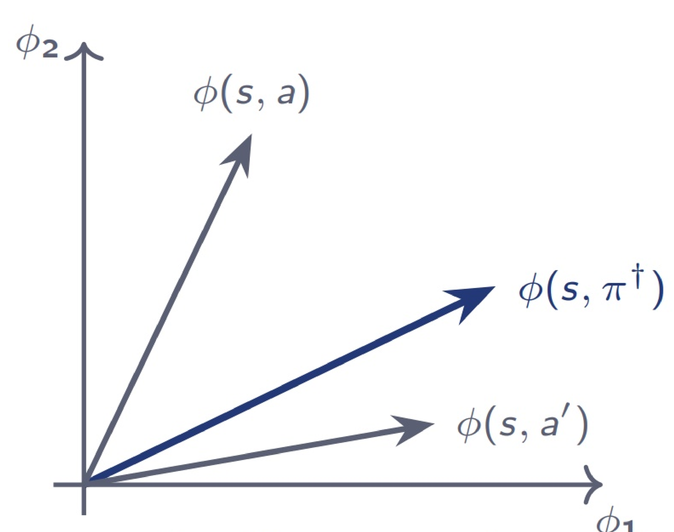
Linear MDP assumption:

MDP = $(\mathcal{S}, \mathcal{A}, H, \{\mathbb{P}_h, r_h\}_{h=1}^H)$

$$\mathbb{P}_h(\cdot \mid s,a) = \sum_{i=1}^{d} \phi_i(s,a)\, \mathbb{P}_h^{(i)}(\cdot).$$

$$r_h(s,a) = \phi(s,a)^\top \theta_h$$

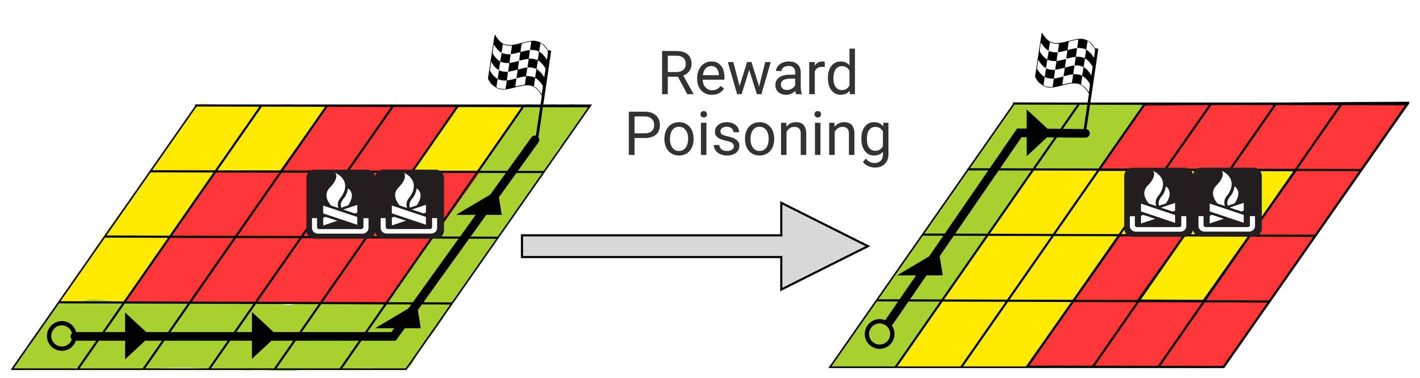
$$\|\theta_h\|_2 \le \sqrt{d}$$



each action ↦ a vector; rewards share one direction θ_h

Note: No linear constraint for mapping $\phi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}^d$

Attackability: an attack strategy costing *o(T)* making any no-regret learner follow adversarial target policy in at least *T - o(T)* episodes with high probability for large *T*.



White vs Black box: White box has access to "hidden" functions and variables of linear MDP while black box must estimate them before-hand. Algorithms in main paper.

Intrinsic Robustness Characterization

Objective: Find reward parameters maximizing reward "gap".

$$\epsilon^* = \max_{\epsilon, \{\theta_h^\dagger\}_{h=1}^H} \epsilon$$

Optimality: Ensure adversarial target actions keep a reward "gap" after rollout.

$$Q_{\{\theta_h^\dagger\}}(s, \pi^\dagger(s)) \ \ge\ \epsilon + Q_{\{\theta_h^\dagger\}}(s,a),$$

Cost Invariance: No poisoning the reward of adversarial target action.

$$\langle \phi(s,\pi^\dagger(s)), \theta_h^\dagger \rangle = \langle \phi(s,\pi^\dagger(s)), \theta_h \rangle,$$

Attack Budget: Poisoning strength limits.

$$\|\theta_h^\dagger\|_2 \ \le\ \sqrt{d}, \forall h.$$

Robustness Test: CQP's "gap" decides intrinsic robustness

$\epsilon^\star \le 0$	$\epsilon^\star > 0$
Robust: No attack can achieve goal.	Attackable: poisoning strategy exists.

Empirical Setting

Environments: HalfCheetah, MountainCart, Pendulum, and synthetic environments.

Attacks: Baseline AT attack v.s. our White/Black box attacks.

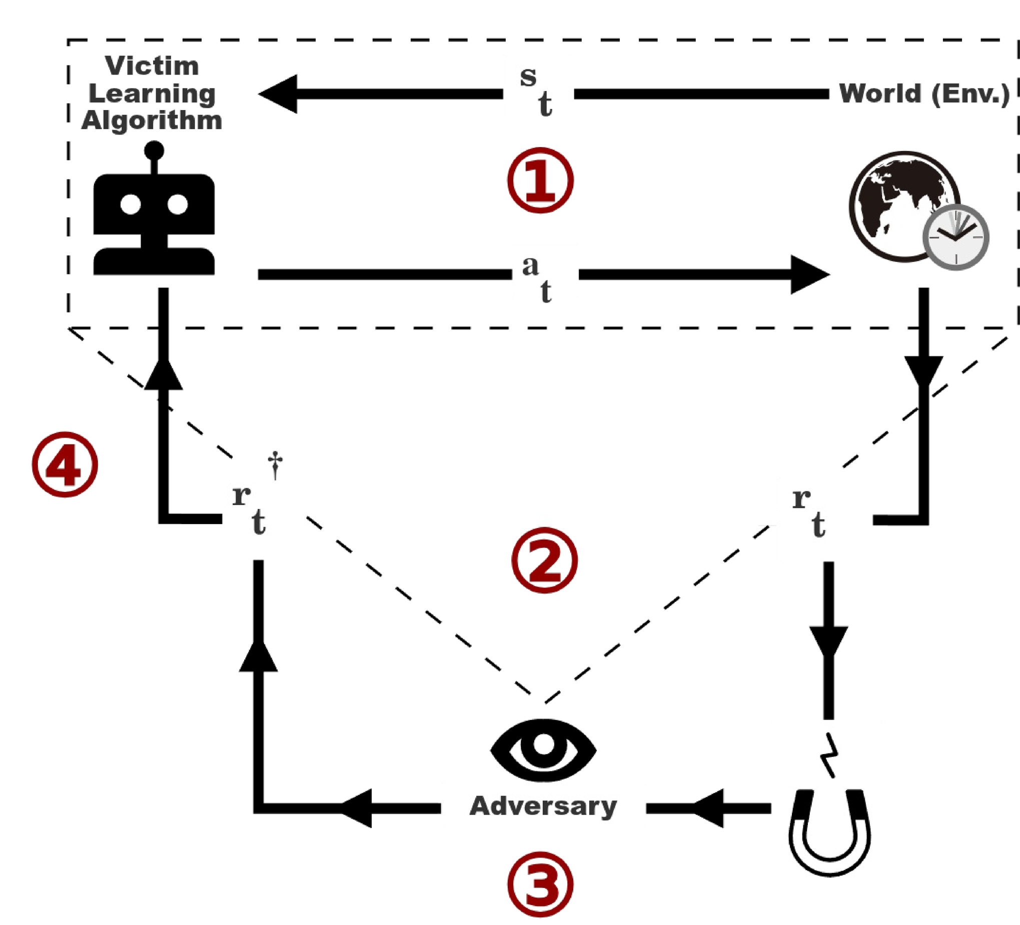
Learner: Non-robust LSVI-UCB

Target Policy: Learned from "hacked" environment. Fixed before every experiment.

HalfCheetah: Run backwards not forward.

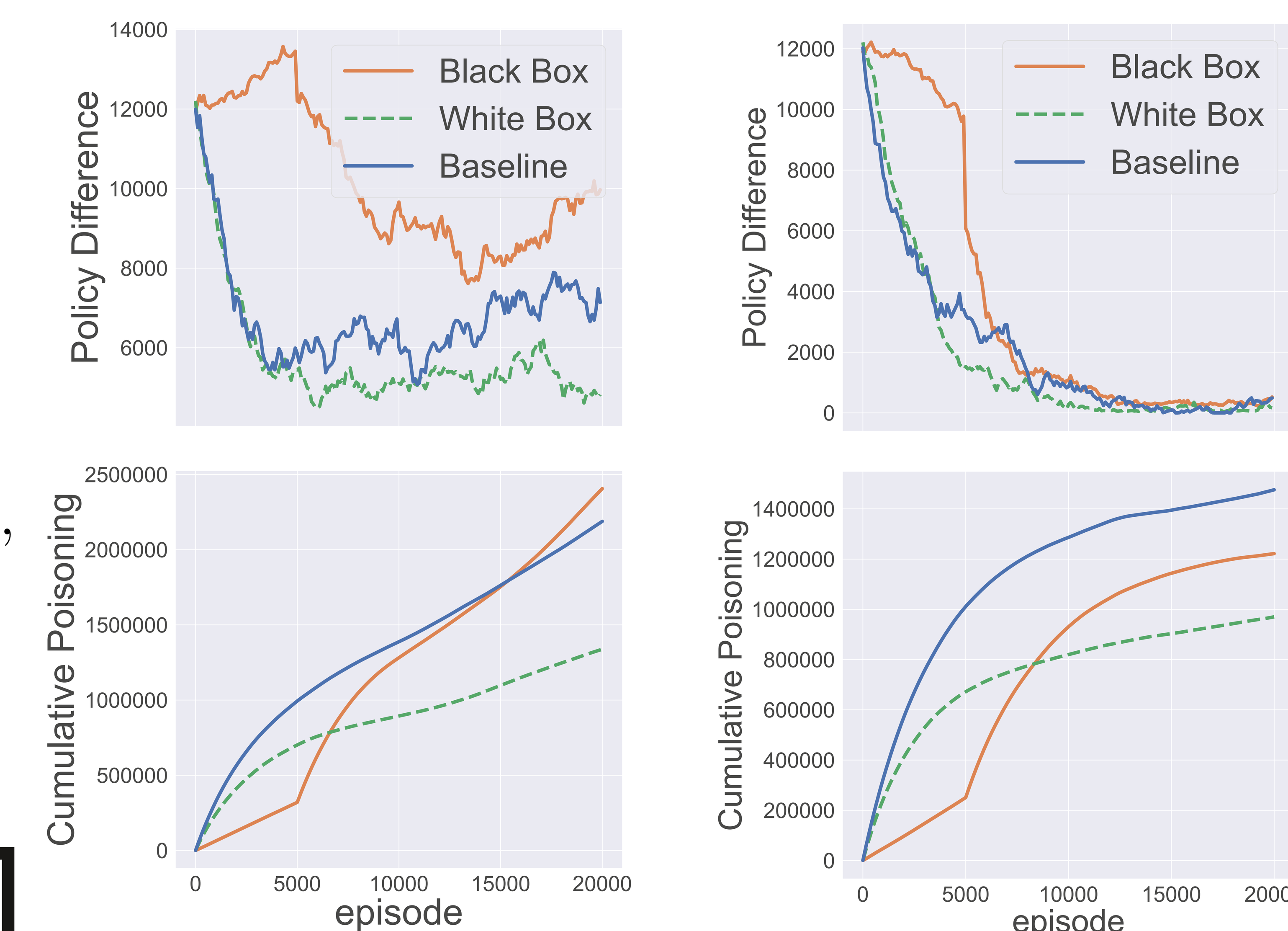
Procedure: Record every timestep the poisoning attacks and policy deviation (policy difference: L2 norm).

We test on 200 environments (100 for each CQP outcome).



(1) Learner gets s_t, selects a_t.
(2) Adversary observes: (s_t, a_t, r_t).
(3) Reward poisoning: $r_t \to r_t^\dagger$.
(4) Learner adjusts policy with $(s_t, a_t, r_t^\dagger)$

CQP Prediction vs Poisoning Behavior



Robust: $\epsilon^\star \le 0$

Policy difference never converges while cumulative costs keep growing.

Vulnerable: $\epsilon^\star > 0$

Policy difference converges towards zero while cumulative costs flatten over time.

Conclusions

Our work makes the following contributions:

1) Intrinsic robustness tied to underlying problem structure.

2) A CQP characterization for robustness certification.

3) Empirical demonstration of CQP's accuracy and theory.

We hope our research will expand the fundamental understanding of adversarial attacks and inspire the design of trustworthy decision making algorithms.